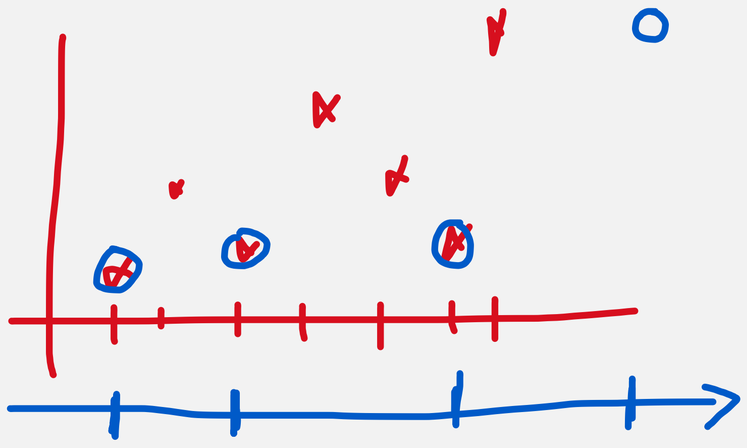
$$e^x \geq 1 + x + \frac{x^2}{2} + \dots + \frac{x^n}{n!}$$

$$e^{\sqrt{x}} - 1 \geq \sqrt{x} \Rightarrow \frac{1}{e^{\sqrt{x}}-1} \leq \frac{1}{\sqrt{x}}$$

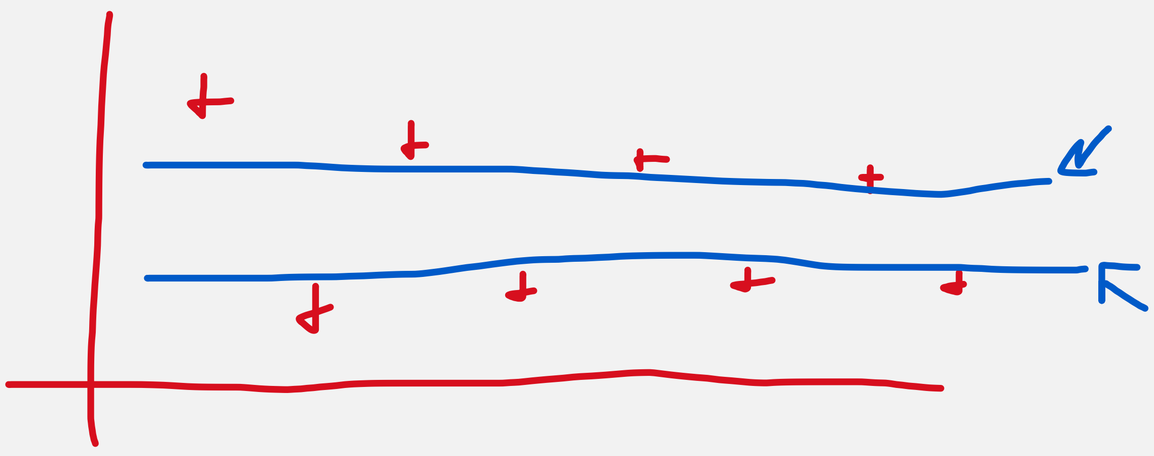
$$\cos x = \sum_{n\geq 0} \frac{(-1)^n x^{2n}}{(2n)!}, \quad e^{x^2} = \sum_{n=0}^{N} a_n x^{2n} + O(x^{2N})$$

(u_n), $(u_{\varphi(n)})$ con $\varphi : \mathbb{N} \to \mathbb{N}$ y $\uparrow$



$(u_0, u_1, \dots, u_n, \dots)$

$(u_0, u_1, u_3, u_{31}, \dots)$

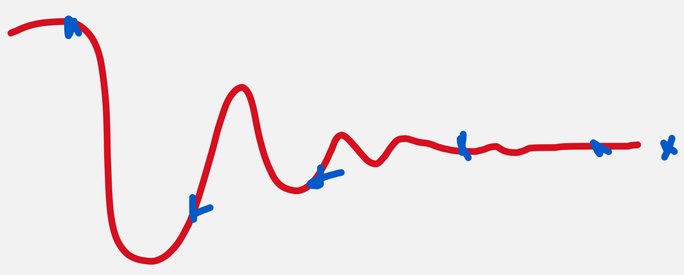


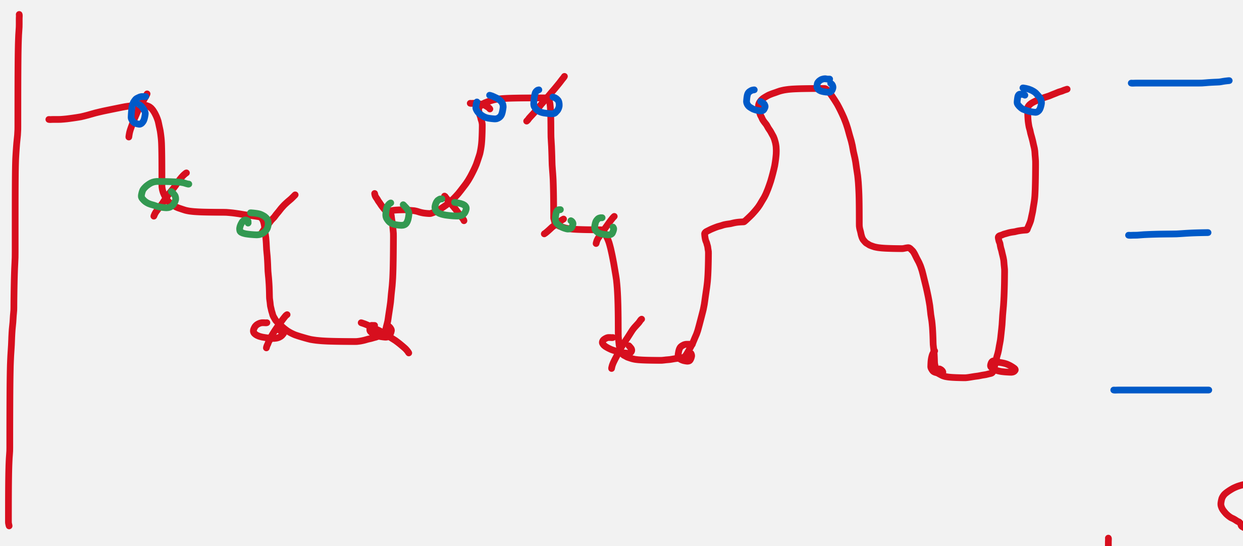
$\exists \varphi : \mathbb{N} \to \mathbb{N}$ $\uparrow$

$u_{\varphi(n)} \to \underline{L}$

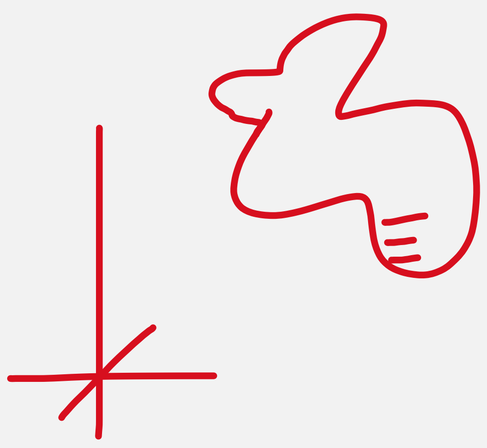
$u_{2n} \to L$

$u_{2n+1} \to L'$









$A \subset \mathbb{R}^d$

$(a_n) \subset A^{\mathbb{N}}$

$a_n \to L$



$\mathbb{R}^2$

$\frac{(x,y)}{\sqrt{x^2+y^2}} < 1$

$a_n : 0, \left(\frac{1}{2}\right), \dots, 0, \dots \to (x,y)$